

OPEN-SOURCE CONTINUOUS-CULTURE BIOREACTORS:

A CORRECTION, A COMPARISON, A TWO-INSTRUMENT PROPOSAL, AND A PATH TO AI-DIRECTED EVOLUTION

Morgan Hough • prepared with AI assistance (Claude, Anthropic)

github.com/m9h/biopunk-bioreactor • CC BY 4.0 • a reader’s audit, not peer review

The thesis: three purposes, one lifecycle

Low-cost open bioreactors are compared as if they compete for one role. They do not — the field serves **three purposes**, stages of a lifecycle, not a ranking:

proto characterise — hold a defined state, measure richly in situ. *Champion: Chi.Bio (an instrument).*

devo evolve — weeks of unattended selection producing *archived, replicated* evidence. *Champion: none yet.*

bior scale out — many reliable reactors, turnkey. *Champions: Pioreactor, eVOLVER.*
proto → **devo** → **bior**: prototype, evolve, produce. A machine optimised for one is measurably worse at another — so build purpose-built instruments and **share the parts**, don’t merge.

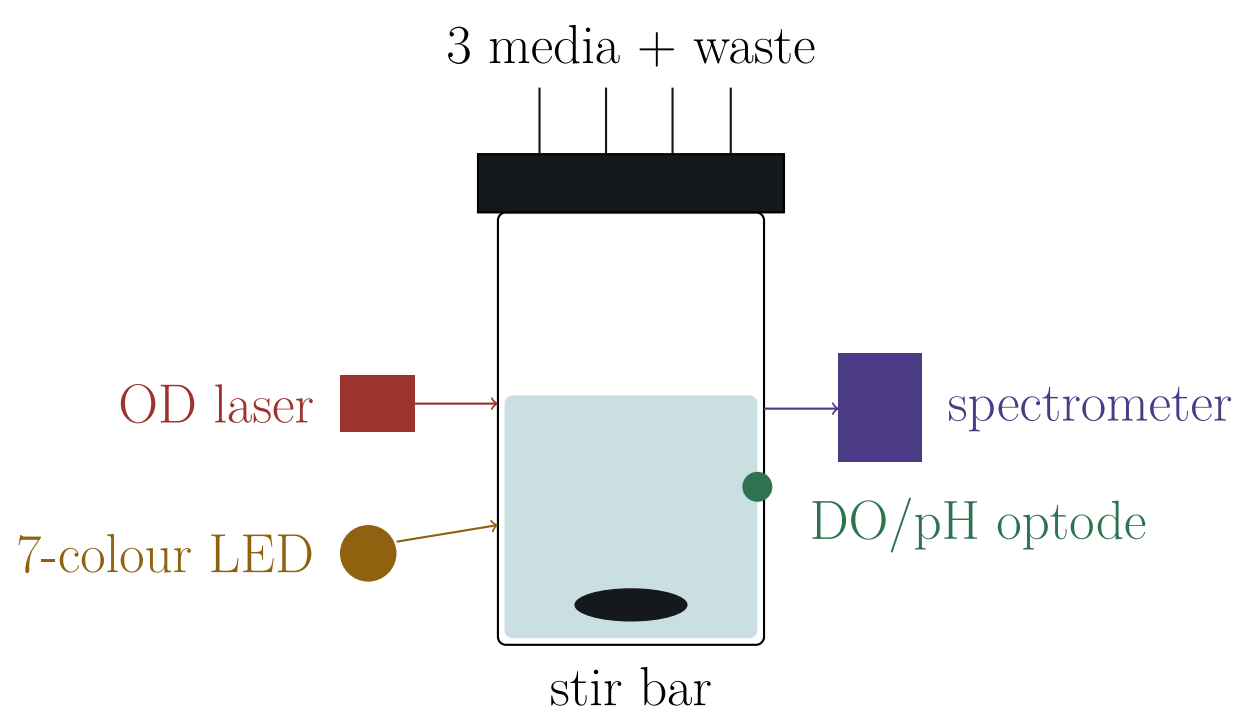
Three lineages, seventy years

Each purpose is its own research tradition — a founding paper and a landmark demonstration:

	Founding lineage	Landmark
proto	chemostat as instrument + closed-loop control (Monod; Novick-Szilard 1950; Miliias-Argeitis 2011)	optogenetic feedback holding expression + growth (2016)
devo	chemostat selection → LTEE → PACE (Novick-Szilard 1950; Lenski 1991; Esvelt 2011)	aerobic citrate use in the LTEE (2012)
bior	continuous culture for production → arrays → eVOLVER (Herbert 1956; Miller 2013; Wong 2018)	eVOLVER ePACE: compact Cas9 (2022)

Now one \$300 device can, in principle, serve all three — and the argument here is that it should not try to at once.

The instrumented reactor (proto)



Non-contact optics read *through the wall* — so tubes hot-swap and stay sterile. **Eight run in parallel** on one computer.

Six platforms, like for like

	Chi.Bio	OpenEvo	Pioreactor	eVOLVER	EVE
Purpose	proto	devo	bior	bior	devo
Paper	yes	yes	none	yes	yes
Parallelism	8	1	cluster	16	3+
Rich sensing	spectro.	OD	plugin	config.	OD
Archiving	no	no	no	no	no
Cost / unit	\$300	\$300	\$329	<\$2k/16	\$115
Build	PCB	guided	buy	shop	mod.

Two patterns: **nobody archives** (the field’s blind spot), and the only paperless platform is the only pure product.

OpenEvo: a fair correction

The recent OpenEvo preprint is a real contribution — cheapest buildable selection cyclor, best manual — but has correctable flaws:

- Error:** abstract “55% faster” vs. results “36% increase” — same data, one mislabelled (7.6 → 4.9 h is +55% rate).
- Error:** claims Pioreactor hardware is closed — it is **CC BY-SA 4.0** open.
- Method:** $n = 1$, no replicates ⇒ no causal attribution; no archiving; default mapper, not **breseq**, for IS-mediated deletions.

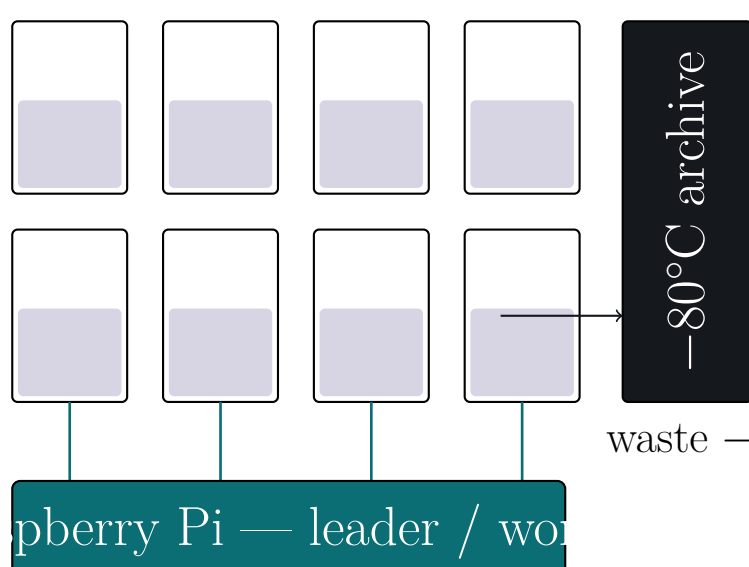
Why devo has no champion

devo is defined by **evidence**, not actuation. OpenEvo actuates but produces no learnable signal. **The apparatus is the champion; a smarter vessel is not.** What the actuator must connect to (only one is new hardware):

- replicate manifold (borrow Pioreactor’s cluster) → convergence
- automated chilled archiver** — the field’s missing freezer → fitness
- sequencing + **breseq** → correct variant calls
- fitness loop (reuses OD hardware); selection design (growth-coupling)

The fleet: eight chambers, one interface (devo / bior)

8 chambers, one interface



Many cheap chambers under one controller, with the **automated chilled archiver** on the waste line — the field’s missing freezer, and what makes fitness measurable.

Three biopunk builds & what they buy

proto — updated Chi.Bio. Chi.Bio core + OpenEvo gas vessel + through-wall DO/pH optodes + C12880MA spectrometer + Pi. *Benefit:* the most complete non-contact in-situ characterisation instrument in open hardware.

devo — evolution + evidence. Pioreactor cluster (or OpenEvo base) + 3-media fluidics + replicate fleet + **archiver** + **breseq** + fitness assay. *Benefit:* fills the empty **devo** slot — first open platform with *publishable* evolution evidence.

bior — Pioreactor + add-ons. Buy the cluster; add the shared optode module (plugin), eVOLVER sleeves only where needed. *Benefit:* reliable parallel cultivation now — mostly bought, not built.

Three configurations of **one shared platform:** Raspberry Pi + one module footprint + one calibration reference.

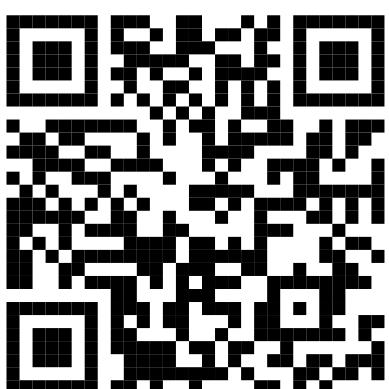
The goal: AI-directed evolution

OpenEvo *pitched* “AI to supervise the evolution” but built only the actuator. **The observation layer an AI needs *is* the evidence apparatus** — so **apparatus = champion, AI = autopilot**; build the apparatus first, then close the loop:

- observe** growth + fitness + genotype (**breseq**)
 - act** on media / light / dilution + **CRISPR-targeted mutation** (EvolvR, an **IGI** tool)
 - learn** rule-based → Bayesian opt → RL, across the fleet
- CRISPR sharpens every layer: EvolvR (targeted diversification), CAMERA (in-cell recorder), Cas-coupled selection. *Prove it: AI-directed vs. fixed schedule, same target.*

Take-aways

- Three purposes, not one contest — different winners.
- Evolution has **no champion**: nobody archives or replicates. *That is the opening.*
- The killer feature nobody has shipped: an **automated chilled archiver**.
- Build the apparatus first; the AI autopilot follows.



github.com/m9h/biopunk-bioreactor

scan for the paper, slides, data & all docs · public · CC BY 4.0